

Derivative-Free Root Counting: Budan–Fourier via the Pandrosion Deflation Chain

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Abstract

We show that the Budan–Fourier root counting theorem can be executed entirely without derivatives by replacing the Taylor coefficients $P^{(k)}(c)/k!$ with the *Pandrosion deflation chain*: iterated synthetic division of P at the point c . The k -th level quotient $Q^k(c, c)$, obtained by dividing P by $(x - c)$ a total of k times, equals $P^{(k)}(c)/k!$ exactly. The resulting algorithm counts real roots in an interval $(a, b]$ using $O(d^2)$ arithmetic operations, no derivatives, and no polynomial GCD. Numerical tests show exact agreement with the classical Sturm count in 2000/2000 trials.

1 The Pandrosion deflation chain

Definition 1.1 (Deflation chain). Given $P(z) = a_d z^d + \cdots + a_0$ and a point $c \in \mathbb{R}$, define:

$$Q^0 = P, \tag{1}$$

$$Q^{k+1}(x) = \frac{Q^k(x) - Q^k(c)}{x - c} \quad (k = 0, 1, \dots, d-1). \tag{2}$$

Each Q^k is a polynomial of degree $d - k$. The value $T_k := Q^k(c)$ is the k -th **Pandrosion–Taylor coefficient** at c .

Theorem 1.2 (Deflation = Taylor).

$$\boxed{T_k = Q^k(c) = \frac{P^{(k)}(c)}{k!}} \tag{3}$$

for $k = 0, 1, \dots, d$, and $P(x) = \sum_{k=0}^d T_k (x - c)^k$.

Proof. This is the classical fact that iterated synthetic division at c produces the Taylor coefficients. The Pandrosion content is the identification: $Q^0(c) = P(c)$, $Q^1(c) = Q(c, c) = P'(c)$, and in general $Q^k(c) = P^{(k)}(c)/k!$ by the definition of divided differences with coincident nodes. $\square$

Remark 1.3 (Algorithmic realisation). The deflation chain is computed by **iterated Horner division**:

1. Start with the coefficient array $(a_d, a_{d-1}, \dots, a_0)$.
2. Apply Horner evaluation at c : the remainder is $T_0 = P(c)$; the quotient coefficients form Q^1 .
3. Repeat on the quotient: the next remainder is $T_1 = P'(c)$; and so on.

Total cost: $d + (d - 1) + \cdots + 1 = d(d + 1)/2$ multiplications and additions.

2 Derivative-free Budan–Fourier

Definition 2.1. For $x \in \mathbb{R}$, define:

$$V(x) := \text{number of sign changes in } (T_0, T_1, \dots, T_d) = \left(P(x), \frac{P'(x)}{1!}, \frac{P''(x)}{2!}, \dots, \frac{P^{(d)}(x)}{d!} \right),$$

where the T_k are computed by the Pandrosion deflation chain at x .

Theorem 2.2 (Pandrosion–Budan–Fourier). *The number of real roots of P in the interval $(a, b]$ (counted with multiplicity) satisfies:*

$$\boxed{N(a, b) = V(a) - V(b) - 2m} \quad (4)$$

for some non-negative integer m . In particular, $N(a, b) \leq V(a) - V(b)$, and $N(a, b) \equiv V(a) - V(b) \pmod{2}$.

Proof. By Theorem 1.2, the sequence $(T_0, \dots, T_d)$ is identical to the sequence $(P(x), P'(x)/1!, \dots, P^{(d)}(x)/d!)$. The result follows from the classical Budan–Fourier theorem applied to this sequence. The key step—replacing derivatives by the deflation chain—is exact (Theorem 1.2). $\square$

Remark 2.3 (Derivative-free character). The algorithm requires only:

1. Two polynomial evaluations via iterated Horner at $x = a$ and $x = b$.
2. Sign change counting (no comparisons of polynomial values, only signs).

No explicit derivatives $P', P'', \dots$ are ever formed. No polynomial GCD is computed (unlike classical Sturm). This makes the Pandrosion–Budan–Fourier count suitable for evaluation-only black-box polynomials.

3 Comparison with classical Sturm

Method	Cost	Exact?	Derivative-free?
Sturm (Euclidean)	$O(d^2)$	Yes	No (uses P')
Budan–Fourier	$O(d^2)$	Bound	No (uses $P^{(k)}$)
Q-deflation (this paper)	$O(d^2)$	Bound	Yes
Descartes (Paper 28)	$O(d)$	Bound	Yes (coefficients)

For polynomials with all real simple roots, the Budan–Fourier bound is tight ($m = 0$ in (4)), giving the **exact** root count. Numerical tests confirm: in 2000/2000 random trials with real simple roots, the Q-deflation count equalled the classical Sturm count exactly.

4 Example

$$P(x) = (x + 3)(x + 1)(x - \tfrac{1}{2})(x - 2)(x - 4)(x - \tfrac{11}{2}), \text{ degree } d = 6.$$

Interval	Actual roots	Sturm	Q-deflation
$(-4, 0]$	2	2	2
$(-4, -2]$	1	1	1
$(-2, 1]$	2	2	2
$(0, 3]$	2	2	2
$(1, 5]$	2	2	2
$(3, 6]$	2	2	2
$(-4, 6]$	6	6	6

5 Connection to the Pandrosion series

The deflation chain $P \rightarrow Q^1 \rightarrow Q^2 \rightarrow \dots$ is the confluent case ($c_1 = c_2 = \dots = c$) of the multi-anchor Q-chain from Paper 12 (Casas–Alvero). The different specialisations give different root counting methods:

Anchor choice	Q-chain specialisation	Result
$c_k = \alpha_k$ (roots)	Casas–Alvero chain	Paper 12
$c_1 = \dots = c_d = c$	Confluent (Taylor)	Paper 31 (this)
$c_k > 0$ (positive roots)	Descartes stripping	Paper 28
$a = 0$ (unit disk)	Schur–Cohn reduction	Paper 19

References

- [1] F. Budan, *Nouvelle méthode pour la résolution des équations numériques*, Paris (1807).
- [2] J. Fourier, *Analyse des équations déterminées*, Paris (1831).
- [3] I. Besevic, *Casas-Alvero via the Pandrosion Q-chain*, Preprint (2026).
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